

Institute of Actuaries of India

Subject SP6 – Financial Derivatives Principles

May 2023 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

The Longstaff-Schwartz approach is a Monte Carlo simulation-based method that uses a regression analysis technique to estimate the expected payoff of the option at each time step. The following are the steps involved:

- a) Define the underlying asset price process: The first step is to define a model for the underlying asset price process. This could be a stochastic process such as Geometric Brownian motion or a jump-diffusion process.
- b) Generate random price paths: Using the model from step 1, generate a large number of random price paths for the underlying asset. These paths should be simulated forward in time, starting from the current price of the asset.
- c) Work backwards in time: For each price path, work backwards in time from the maturity of the option, calculating the expected value of the option at each previous point in time.
- d) Fit a regression model: At each time step, fit a regression model to the simulated price paths and the corresponding expected option values. This regression model estimates the expected option value as a function of the simulated asset prices and other relevant variables.
- e) Use the regression model to estimate the expected option value: Use the estimated regression model to calculate the expected option value at each previous point in time.
- f) Compare expected option value to immediate payoff: At each previous point in time, compare the expected option value to the immediate payoff of exercising the option. Take the maximum of these two values as the estimated option value at that time.
- g) Estimating the optimum exercise time for each path: the time at which the option value is maximum for each path is now identified.
- h) Estimating the price of the option for each path: generate the cashflow at the optimum exercise time for each path by simply exercising the option. Arrive at the value of the option for each path by discounting the cash flow to present value.
- i) Calculate the expected option value: Calculate the expected option value as the average of the estimated option values across all price paths.
- j) Repeat steps 2 to 9 multiple times: To improve the accuracy of the estimate, repeat steps 2 to 9 multiple times and take the average of the resulting present values.

[9 Marks]

Solution 2

- i) The three approaches are
 - Direct transaction in market
 - Using future and forward initially and slowly move to desired exposure by selling and buying equities
 - Use rolling future to maintain the desire exposure.

Direct transaction company can get into specific equities company want immediately, There would be a higher cost bid offer for large immediate transaction as timing may not be appropriate. This does give opportunity for active fund manager to choose under-price security immediately hence benefit from future market movement over and above the index. This would also shield company from any adverse currency movements.

The second approach is transitional measure of having forward or future exposure to index and slowly move out of current exposure to desired exposure. This would reduce the market timing impact of selling and buying but would also reduce probability of fund manager beating the index hence resulting in slightly lower performance fees. There would also be basis risk between underlying asset and index and further potential of adverse FX movement.

Third approach is to have rolling forward and future for longer period which would have been a good option if fund was passive management fund. This would not be appropriate for current fund.

[4]

- ii) The choice of approach is option 1 and option 2. Ideally, we should go for option 2 given this option provide balance of all the factor and period could be chosen to reduce the loss of alpha return.

[2]

- iii) The following changes are expected in the overall portfolio

	Current	Future	Current (Cr)	Value in '000 domestic currency	Future exposure '000 (in domestic currency)	Exposure change
Indian equity	40%	60%	40	400000	600000	200000
UK equity	30%	20%	30	3000	2000	-1000
US equity	30%	20%	30	3750	2500	-1250

	Exposure change (in '000 local currency)	Price of index	Interest annually compounding	Price of future	Number of Future in (LC/Future P)
India (BSE)	2,00,000	60000	8%	61166	3,269.00
UK equity	- 1,000	7000	5%	7086	(141.00)
US equity	- 1,250	4000	6%	4059	(307.00)

[4]

- iv) Profit in scenario 1 with immediate sale

		India index	2%	US index	2%		
FXUK3M	98	India Alpha	4%	US alpha	0%		
FXUS3M	78	UK index	0	Transaction cost	1%		
Portfolio		UK alpha	1%				
	Exposure before (INR) '000	Transaction amount '000 INR	Transaction cost '000 INR	Portfolio after '000 INR	In local currency	Growth (Index + alpha)	Portfolio in INR '000
Indian Equity	4,00,000.00	2,00,000.00	2,000.00	5,98,000.00	5,98,000.00	6,33,880.00	6,33,880.00
UK Equity	3,00,000.00	- 1,00,000.00	1,000.00	1,99,000.00	1,990.00	2,009.90	1,96,970.20
US equity	3,00,000.00	- 1,00,000.00	1,000.00	1,99,000.00	2,487.50	2,537.25	1,97,905.50
							10,28,755.70

[5]

- v)

	Exposure before (INR) '000	In local currency '000	Return '000	In local currency '000 INR	
Indian Equity	400000	400000	424000	424000	
UK Equity	300000	3000	3030	296940	

US equity	300000	3750	3825	298350		
				1019290		
Profit loss on Forward	On expiry value of forward would be equal to value of Index					
	Price	Index now	Profit loss per contract	No of contract	In local currency	In INR '000
Indian forward	61166	61200	34	3,269.00	1,11,146.00	111.15
UK forward	7086	7000	-86	(141.00)	12,126.00	1,188.35
US forward	4059	4080	21	(307.00)	(6,447.00)	(502.87)
						796.63
					Total loss in using option	8,669.07

[5]

[20 Marks]**Solution 3:**

- i) Convexity of a Bond is a measure that shows the relationship between bond price and Bond yield, i.e., the change in the duration of the bond due to a change in the rate of interest, which helps a risk management tool to measure and manage the portfolio's exposure to interest rate risk and risk of loss of expectation. Convexity would be a more appropriate measure of interest rate risk than modified duration in a scenario where interest rate changes are expected to be large and non-parallel across the yield curve, or when the bond has a longer maturity or a larger coupon rate. **[3]**
- ii) Price volatility increases as convexity increases, meaning that a bond with higher convexity will have a greater change in price for a given change in interest rates. **[2]**
- iii) Convexity can be used to hedge against interest rate risk by creating a portfolio of bonds with different convexities, such that the convexity of the portfolio is matched to the desired level of interest rate risk. **[2]**
- iv)

Change of yield	2%			
	Institution 1		Institution 2	
	Asset	Liability	Asset	Liability
Duration impact	-12%	-24%	-18%	-6%
MV of asset	88	76	82	94
	Institution 1		Institution 2	
	Before stress	After Stress	Before stress	After Stress
Net Asset	0	12	0	-12
Total	0			
Change of yield	-2%			
	Institution 1		Institution 2	
	Asset	Liability	Asset	Liability
Duration impact	12%	24%	18%	6%
MV of asset	112	124	118	106
	Institution 1		Institution 2	

	Before	After	Before	After
Net Asset	0	-12	0	12
Total	0			

[4]

v)

Change of yield	2%			
	Institution 1		Institution 2	
	Asset	Liability	Asset	Liability
Duration impact	-12%	-24%	-18%	-6%
Convexity impact	2.0%	4.0%	2.0%	1.0%
MV of asset	90	80	84	95
	Institution 1		Institution 2	
	Before	After	Before	After
Net Asset	0	10	0	-11
Total	-1			
Change of yield	-2%			
	Institution 1		Institution 2	
	Asset	Liability	Asset	Liability
Duration impact	12%	24%	18%	6%
Convexity impact	2.0%	4.0%	2.0%	1.0%
MV of asset	114	128	120	107
	insurance Company		Bank	
	Before	After	Before	After
Net Asset	0	-14	0	13
Total	-1			

As per the convexity approach company would ensure loss of 1 Cr in both rate up and rate down scenario. Convexity approach gives a different conclusion and is better due to larger change of interest rate.

[4]

[15 Marks]

Solution 4:

- i) The CIR (Cox-Ingersoll-Ross) model is a variation of the Vasicek model that allows for time-varying volatility. The model assumes that interest rates cannot go negative, and that volatility is proportional to the square root of the interest rate level. The CIR model is commonly used to model the term structure of interest rates and to value interest rate options.

The Hull-White model extends the Vasicek and CIR models by incorporating a stochastic term structure of interest rates. The model assumes that the short-term interest rate follows a mean-reverting process, while the long-term interest rate follows a separate mean-reverting process that is driven by a stochastic factor. The Hull-White model is often used to model the dynamics of yield curves and to value interest rate derivatives that depend on the shape of the yield curve.

[3]

- ii) CIR model assumes that interest rates follow a process that allows for both mean reversion and volatility clustering, making it better suited for modeling interest rate movements in the short

term. This feature makes the CIR model more appropriate for valuing caps and floors, which are generally options on short-term interest rates.

The Hull-White model allows for a flexible term structure that can be calibrated to match the observed term structure of interest rates. This feature is essential for pricing complex derivatives that depend on interest rates over different time horizons. It allows for volatility to be modeled as a function of time, which makes it more accurate in capturing changes in volatility over time. This feature is crucial in pricing complex derivatives that are sensitive to changes in volatility. It can be extended to a multi-factor model, which allows for more complex and realistic interest rate dynamics. This extension is essential for pricing complex derivatives that depend on multiple interest rate factors.

[5]

[8 Marks]

Solution 5:

- i) A synthetic CDO (collateralized debt obligation) is a type of derivative that is created by pooling together credit default swaps (CDS) or other types of credit derivatives. It is a complex financial instrument that allows investors to bet on the performance of a portfolio of underlying credit assets, such as bonds or loans. The synthetic CDO is divided into different tranches, each with its own level of risk and potential return. The returns from the CDO are derived from the premiums paid on the CDS contracts.

In contrast, a CLO (collateralized loan obligation) is a type of structured credit product that pools together a diversified portfolio of loans made to corporate borrowers. The loans are typically rated below investment grade and are often used to finance leveraged buyouts or other corporate transactions. Like synthetic CDOs, CLOs are divided into different tranches with varying levels of risk and return.

One key difference between synthetic CDOs and CLOs is the underlying assets. Synthetic CDOs are based on credit derivatives, while CLOs are based on loans made to corporate borrowers.

[3]

- ii) The attachment point in a synthetic CDO is an important concept for investors because it determines the level of risk associated with the investment. The attachment point represents the threshold at which losses on the underlying assets will begin to affect the investors in a particular tranche of the CDO.

For example, suppose a synthetic CDO has three tranches: senior, mezzanine, and junior. The senior tranche has the lowest risk and is the first to receive payments from the underlying assets. The junior tranche has the highest risk and is the last to receive payments. The attachment point for the junior tranche is the level of losses on the underlying assets that must be reached before the junior tranche begins to experience losses.

Investors in the junior tranche of a synthetic CDO typically receive higher returns than investors in the senior or mezzanine tranches because they are taking on more risk. However, the attachment point is an important consideration because it determines the potential losses that investors may incur. If the underlying assets perform poorly and losses exceed the attachment point for the junior tranche, investors in that tranche will experience losses.

Therefore, investors in a synthetic CDO need to carefully evaluate the attachment point for each tranche and consider how it fits with their risk tolerance and investment objectives. If an investor is willing to accept a higher level of risk in exchange for potentially higher returns, they may be interested in investing in the junior tranche with a lower attachment point. However, if an investor is more risk-averse, they may prefer to invest in the senior or mezzanine tranches with higher attachment points and lower potential losses.

[3]

[6 Marks]

Solution 6:

- i) Market risk is the risk that the value of a portfolio will fall due to an adverse change in the level or volatility of the market price of interest-rate instruments, currencies, commodities and equities.

The basic market risks in the portfolio are:

- a. **Directional price risk, or delta:** This is the change in the value of the portfolio due to changes in the prices of the underlying instruments — all other variables remaining constant.
 For linear derivatives like futures, cash bonds and swaps, delta is a reasonably accurate forecast of market risk. (In fact, linearity does not exactly apply to ordinary bonds and swaps, but the error in assuming linearity is small.)
 Delta (in conjunction with gamma and vega) also works well for vanilla options on futures and forwards, and delta hedging is a fairly straightforward technique.
 Delta produces less reliable forecasts of market risk for exotic options, where the payoff can be very non-linear.
- b. **Convexity risk, or gamma:** This is the change in the delta arising from changes in the price of the underlying instruments — all other variables remaining constant.
 Vanilla options have well behaved gamma profiles, so only at expiry can there be significant risk.
 Exotic options have very variable gamma and vega (volatility) profiles so risk profile at the current market level is misleading. Some exotics are virtually riskless at current levels, but “blow up” somewhere far away (e.g. barrier options or cancellation options).
 Stress tests or jump diffusion models need to supplement the “greeks” here.
- c. **Volatility risk, or vega:** This is the change in the value of the portfolio arising from changes in implied volatility of the underlying instrument — all other variables remaining constant.
 Vega risk can only be hedged by taking positions in options (often traded options — but then there can be basis risk as described below).
- d. **Basis risk:** This is the change in the value of the portfolio arising from changes in correlated variables.
 This is particularly important with options on futures versus options on cash markets or indices, which can get out of line with each other before expiry.
 Hedged portfolios often experience basis risk, e.g. hedging the interest rate risk from a long position in a corporate bond using a government bond runs the risk.
- e. **Time-decay risk or theta:** This is the change in value of the portfolio arising from the passage of time — all other variables remaining constant.
 Time decay is not usually hedged directly but treated as an “accrual” cost of a portfolio.
 Instead, gamma and vega are hedged and there is a net effect on theta.
- f. **Interest rate risk or rho:** This is the change in the value of the portfolio arising from changes in interest rates used to discount future cash flows — all other variables remaining constant.
 For currency options, which are based on currency pairs, there are two interest rates involved.

Other more minor issues relating to market risk are:

Structured bonds can mostly be decomposed into more basic securities, usually with a bond underlying. For example, an inverse floater can be transformed into a FRN plus twice

the amount of a fixed-floating interest-rate swap. If there is significant optionality, however, this approach will not be adequate.

Credit spread risk is usually treated as a type of market risk present to measure the sensitivity to default risk in non-government bonds.

Liquidity risk is often linked with market risk — this relates to the difficulties of closing out positions in times of market stress.

And finally, Reset risk for cash flow resets in swaps or dividend estimation for equity indices.

[10]

- ii) Interest-rate-derivatives traders (including bond traders) want to be able to quote prices which are in line with prices being quoted by other traders. However, the use of the Vasicek model may violate the no – arbitrage condition, as it may not be exactly in line with the full market prices.

The Hull & White (HW) is better in that it allows the trader to price interest-rate linked contracts more accurately by reflecting the current *observed* term-structure of interest rates. Also, the HW can be easily extended to include a time-varying but deterministic $\sigma(t)$, which allows the model to be calibrated for all traded option prices as well as zero-coupon bond prices and so use the entire yield **and** volatility curves.

However, a disadvantage of this version of HW is that model calibration is much more difficult as compared to the Vasicek or simple HW. And even this version of HW, with its extra flexibility, cannot cope with two- or three-factor problems (options based on more than one asset).

Otherwise, Vasicek and HW are very similar. Since $\mu(t)$ is deterministic, the simple HW model is just as easy to set up and manipulate as the Vasicek model. But both HW and Vasicek suffer from the same drawback that interest rates might become negative (although the time-dependent parameters in HW can make this very unlikely). Also, it needs to be noted that not all types of options are covered by either model as described above, and models may break down under stress scenarios. (Stress testing is very important for complex option portfolios where financial effects are really only seen in the tails of the distribution. In stress situations, bid-offer spreads can widen, liquidity can dry up and firms can face extraordinarily large cash collateral demands.)

[5]

[15 Marks]

Solution 7:

- i) Rewriting the given equation, we get the annual marginal probability of default given by

$$PD = \frac{1.5\%}{(1 - 0.50)} = 3.00\%$$

Hence, the cumulative PD for five years is

$$\begin{aligned} d + s_1d + s_2d + s_3d + s_4d = \\ 3.00\% (1 + 0.970 + 0.941 + 0.913 + 0.885) = \\ 14.1\% \end{aligned}$$

Hence,

$$\begin{aligned} s_1 &= (1 - 3.00\%) = 0.970 \\ s_2 &= s_1(1 - 3.00\%) = 0.941 \\ s_3 &= s_2(1 - 3.00\%) = 0.913 \\ s_4 &= s_3(1 - 3.00\%) = 0.885 \end{aligned}$$

The expected number of defaults is thus $100 * 14.1\%$ or approximately 14.

[5]

- ii) With a recovery rate of 50%, the expected loss is therefore 7% of the notional. Thus, all tranches up to the 7% point are wiped out.

[2]

[7 Marks]

Solution 8:

- i) Moneyness is the relative position of the current price (or future price) of an underlying asset (e.g. stock) with respect to the strike price of a derivative. First, we estimate the forward swap rate for duration 15 to 20 for the given forward bond prices. In this case, it is:

$$\frac{\text{Bond Price (15)} - \text{Bond Price (20)}}{\text{Sum (Bond Price from 15 - 20)}} = 4.65\%.$$

Thus, the product is in the money.

[3]

- ii) We calculate the risk – free rate at time $t = 15$ from the bond price to be equal to 2.9%. We can then use Black's formula.

$$P_{PS} = NA[i_F \Phi(d_1) - i_S \Phi(d_2)]$$

Where,

$$A = \sum_{i=1}^n e^{-it_i}(t_i - t_{i-1}), \quad d_1 = \frac{\ln \frac{i_F}{i_S} + \frac{1}{2} \sigma_F^2 T}{\sigma_F \sqrt{T}}, \quad d_2 = \frac{\ln \frac{i_F}{i_S} - \frac{1}{2} \sigma_F^2 T}{\sigma_F \sqrt{T}} = d_1 - \sigma_F \sqrt{T}$$

Using the estimate of the market volatility to be 25%, we get:

Intermediate Calculations:

Swap Tenor (years):	5
Swap Rate Period (years):	15
D_t (Volatility * sqrt(15)):	0.9682458
d_1 :	0.4826274
d_2 :	- 0.4856185
$\Phi(d_1)$:	0.6853198
$\Phi(d_2)$:	0.3136188
A:	4.4525463
Number of payments/ year:	1
Principal Amount:	\$ 10 billion
15 year forward swap rate:	4.00%
Volatility of the forward rate:	25.00%
Risk Free rate:	2.90%
Price:	\$ 420 million

[4]

- iii) Use the implied volatility estimated from the earlier part as 25%. The new forward rate can be estimated from the bond price which is equal to 7% and the new risk free interest rate is 3.5%. The new price can again be calculated using Black's model

Dt (Volatility * sqrt(15)):	0.9682458
d1:	1.002194
d2:	0.0339488

The new price is \$85 million.

[5]

- iv) Market is given higher price than that can be estimated from implied volatility as the implied volatility is not the same when interest rates go up. This is referred to as the volatility smile where the underlying implied volatility changes with the moneyness of the option.

[3]

[15 Marks]

Solution 9:

The following can be the implications:

- The market has been witnessing a good growth in volumes indicating that the market depth is increasing and more players are probably entering the market. The increase in STT rate may also hence get absorbed without many disruptions.
- Since the STT is on contract value for futures but on premium for options there maybe a swing from futures volume to options volume. A good derivative market requires healthy volume in futures as well as options. This hike may lead to distortions making futures more unattractive compared to options which is not good for the market.
- Market depth is mainly provided by day traders, arbitrageurs, high frequency trading algorithms etc. Increase in STT can lead to higher breakeven levels for these players. The trading volumes may hence shrink especially when the market volatility decreases. Many players may even exit the market. The STT collection target of the government may also not be achieved if volumes shrink.
- Investors and hedgers who take positional bets with low trading frequency maybe able to absorb the impact more easily. However, if the trading volumes shrink then the bid-ask spread may increase and the cost of hedging or the cost of entering or exiting a position may increase due to the higher spread.
- If there are competing exchanges abroad where the transaction costs are lower and they offer derivatives on the same underlying then the players may shift their trades to those exchanges

[5 Marks]
